

Latent Diffusion for Synthetic Brain MRI: An Exploratory Evaluation of Fidelity, Memorisation, and Segmentation Utility

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Abstract

Synthetic brain MRI could expand scarce labelled datasets, but realistic-looking images alone do not show that a generation pipeline is viable. An exploratory assessment must also ask whether samples resemble held-out data without copying training images, retain diversity, respond to unseen tumour masks, and show at least conditional downstream utility. We do not propose a new architecture. Instead, we evaluate a conventional mask-conditioned 2-D latent diffusion model (LDM) on BraTS 2020 using patient-level splits, validation-based early stopping, a nearest-neighbour memorisation check, held-out fidelity and diversity metrics, unseen-patient masks, and segmentation as a secondary stress test. The evidence is mixed. Without regularisation, FID continues to improve while the model memorises: by epoch 200, 97 % of samples lie closer to a training slice than real held-out slices do. Cached affine augmentation postpones this failure and yields the best early-stopped model (validation FID 24.7 vs 29.9). At 256 px, the selected model reaches FID 10.45 against a 9.55 floor between two real sets and generates from unseen tumour masks with little loss in fidelity, providing encouraging distributional evidence within this dataset. Practical utility is conditional. Mixing synthetic slices into segmenter training lowers mean Dice by 0.030–0.036 at 128 px, whereas synthetic pre-training gains +0.021 / +0.014 over a compute-matched control at 10 % / 25 % real data, and 256 px mixing gains +0.027 with 26 real patients. Autoencoder controls are consistent with the latent bottleneck being the principal limitation at 128 px. Within this 2-D BraTS setting, latent diffusion is therefore a plausible research pipeline, but its viability depends on resolution, regularisation, checkpoint selection and how the samples are used; clinical realism and broad utility are not established.

1 Introduction

Synthetic images are an attractive response to the scarcity of labelled medical data, and denoising diffusion models [14, 24] have produced increasingly realistic brain MRI samples [17, 18, 23]. The practical question, however, is not simply whether a sample looks convincing. For this exploratory study, *viability* means that an established generation pipeline can be trained reproducibly, produces samples that resemble held-out data while remaining diverse and distinct from the training set, responds to unseen tumour masks, and shows enough conditional utility to justify further investigation. These criteria are deliberately narrower than anatomical validity, diagnostic usefulness or clinical readiness, none of which is evaluated here.

We do not introduce a new generator architecture or optimisation method. Instead, we use a conventional mask-conditioned LDM to map the evidence for and limits of synthetic brain MRI generation on BraTS 2020. The objective is exploratory, while the evaluation rules are pre-declared to reduce cherry-picking: patient-level splits, validation-based checkpoint selection, a flip-aware nearest-neighbour memorisation check, model selection without test data, and three seeds with patient-matched pairs for the secondary segmentation assessment. The study asks four questions:

- How do training length, regularisation and checkpoint selection affect fidelity and memorisation (Sect. 5.1)?
- Do selected models produce diverse samples that resemble held-out data and respond to tumour masks from unseen patients (Sect. 5.2)?
- How do image resolution and the pretrained autoencoder constrain the evidence for viability (Sect. 5.5, 5.6)?
- As a secondary test, when do synthetic image–mask pairs help or hurt downstream segmentation (Sect. 5.3–5.4)?

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The paper’s contribution is therefore an empirical assessment of feasibility and boundary conditions rather than a claim of architectural novelty.

2 Related work

Diffusion models for medical image synthesis. Pinaya et al. [23] trained a 3-D LDM on 31,740 UK Biobank brains; Khader et al. [18] and Müller-Franzes et al. [22] compared diffusion models with GANs on several modalities and found diffusion samples more realistic. Fernandez et al. [9] trained segmenters on fully synthetic brain data and reported near-parity with real data for some tasks. Surveys [7, 17] list many more. Most of this work reports fidelity metrics or a single downstream run; few report a memorisation check or a control for the autoencoder. We therefore treat the established LDM pipeline as the object of evaluation and ask how much evidence its outputs provide for viability in one controlled setting.

Memorisation. Carlini et al. [6] and Somepalli et al. [26] showed that diffusion models reproduce training images, and Gu et al. [11] related the effect to data size and training length. In medical imaging, Dar et al. [8] found that a 3-D LDM memorised a large share of its training volumes, and Akbar et al. [1] reported the same for 2-D brain MRI and chest X-rays, both with nearest-neighbour distances as we do. Data augmentation is the classical remedy for generator overfitting [16]; we apply it in latent space so that the frozen autoencoder is run once per variant.

Synthetic data for segmentation. GAN-based augmentation helped classification in early work [10], but gains for segmentation are mixed and depend on the ratio of synthetic to real data and on how the synthetic labels are obtained. We use the conditioning mask as the label, restrict each low-data segmenter to synthetic slices conditioned on masks of its own patients, and test three uses of the same pool (mixing, a 1:1 ratio, pre-training).

3 Data and protocol

Data. The BraTS 2020 training set [2, 3, 21] contains 369 patients with co-registered, skull-stripped FLAIR, T1, T1ce and T2 volumes ($240 \times 240 \times 155$ at 1 mm^3) and expert labels (necrotic core, oedema, enhancing tumour). Patients are split once, at patient level, into 258 training, 37 validation and 74 test patients (seed 0; the split file is committed). Each modality of each volume is clipped to its [0.5, 99.5] intensity percentiles inside the brain and rescaled to [0, 1]; axial slices whose tumour covers at least 0.1% of the slice are kept, centre-cropped to 224×224 (the brain never extends beyond that window) and resized to 128 or 256 px. This gives 15,895 / 2,159 / 4,623 tumour-bearing slices for train / val / test. FLAIR, T1ce and T2 are mapped to the three input channels of the pretrained autoencoder. Evaluation regions follow the challenge: whole tumour (WT), tumour core (TC) and enhancing tumour (ET).

Pre-declared protocol. Every analysis below was written into the repository’s protocol file before its results were seen; the timestamps and the exact reading rules are in the released `docs/EXPERIMENTS.md`. In particular the checkpoint rule, the model-selection rule, the segmentation protocol, its three secondary analyses, the compute-matched control, the 256 px repeat and the decoder fine-tune study (with the condition under which it would stop early) were fixed in advance. All numbers come from run directories that record the resolved configuration, the git commit and the metrics.

4 Methods

Exploratory evaluation framework. We assess viability across five linked dimensions: technical feasibility at 128 and 256 px; distributional fidelity and diversity relative to held-out real slices; non-memorisation under checkpoint selection; conditional generation from masks of unseen patients; and downstream segmentation as a secondary test of practical utility. No single metric is treated as decisive. In particular, FID/KID and sample grids provide distributional and visual evidence, not a radiologist assessment or evidence of clinical validity.

4.1 Latent diffusion model

We follow Rombach et al. [24]. The frozen Stable Diffusion autoencoder `sd-vae-ft-mse` compresses a slice by $8\times$ into a 4-channel latent ($16 \times 16 \times 4$ at 128 px, $32 \times 32 \times 4$ at 256 px). A U-Net (channels 128/256/512/512, two residual blocks per level, self-attention at the two coarsest levels, 102 M parameters) predicts the noise ϵ

under a 1,000-step linear schedule [14]. Training uses AdamW [19] ($\text{lr } 10^{-4}$, weight decay 0.01), 500 warm-up steps then cosine decay, gradient clipping at 1.0, an EMA of the weights (0.9999) that is used for every reported sample, and bf16 autocast. Batch size is 64 at 128px and 32 at 256px; 200 epochs correspond to 49.6k and 99.2k steps and take 1.5 h and 2.8 h on one RTX A6000.

Mask conditioning. The four-class mask is one-hot encoded, average-pooled to the latent grid (soft per-cell class fractions) and concatenated to the noisy latent, together with a fifth *null* channel that marks a dropped condition so that “no condition” is distinct from “all background”. Conditions are dropped with probability 0.1 during training, which enables classifier-free guidance [13] at sampling time. Samples are drawn with DDIM [27], 50 steps, $\eta = 0$, guidance scale 2 unless stated.

Cached latents and augmentation. Because the autoencoder is frozen, the posterior of every slice is encoded once and cached (mean and log-variance, sampled on each access, for the slice and its horizontal flip). The regularised recipe (“reg”) adds U-Net dropout 0.1 and six extra cached variants per training slice, each a random affine transform (flip with $p = 0.5$, shift up to 6%, rotation up to $\pm 10^\circ$, isotropic scale 0.9–1.1; zero padding). Each access draws one of the eight variants and applies the identical transform to the conditioning mask. Validation latents are never augmented, so validation losses remain comparable across recipes.

4.2 Checkpoint selection and memorisation check

The validation diffusion loss is computed with fixed noise and evenly spaced timesteps so that epochs are comparable; the EMA weights of the epoch with the lowest validation loss are the reported model. To see what later checkpoints buy, every tenth epoch is scored with the same 2,000 masks and seeds: FID/KID [4, 12] against the real validation slices and a memorisation statistic. For the latter, each sample is compared with every training slice *in both orientations* (the models train with flips, so a copy of a mirrored slice is a copy) by L_2 distance on 64×64 downsampled images; the same distances are computed for real held-out slices, and the *memorised fraction* is the share of samples closer to a training slice than 95% of real held-out slices are. A model that generalises scores about 0.05 by construction.

Model selection. Three 128px mask-conditioned candidates differ only in regularisation: flips only (baseline), plus dropout 0.1, and the reg recipe. The rule, fixed before the runs finished, takes each candidate at its best-validation-loss checkpoint, excludes candidates with memorised fraction above 0.15, and picks the lowest FID against the validation slices. The test set plays no part. The chosen recipe was then reused unchanged for an unconditional 128px model and for the 256px model.

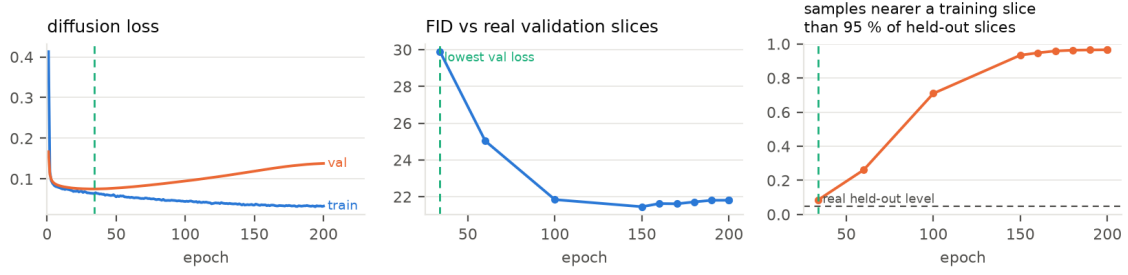
4.3 Distributional fidelity, diversity, and unseen-mask generation

Against all 4,623 real test slices we report FID and KID (InceptionV3 features via torch-fidelity) for the three-channel image and for each modality with the grey channel replicated to RGB, using 5,000 samples per setting; the FID between the real validation and real test slices (10.49 at 128px, 9.55 at 256px) is the floor these numbers should be read against. Diversity is the mean SSIM [28] over 2,000 random sample pairs (real test pairs: 0.652 at 128px, 0.690 at 256px). The autoencoder’s *ceiling*, PSNR / SSIM / LPIPS [29] of $\text{decode}(\text{encode}(x))$ on real test slices, bounds every latent model. Each mask-conditioned model also generates 5,000 samples from masks of *validation* patients, tumour shapes it never saw.

4.4 Secondary assessment: downstream segmentation utility

A 2-D U-Net [25] from MONAI [5] (channels 32–512, two residual units per level, instance normalisation) is trained with Dice + cross-entropy, AdamW (3×10^{-4} , weight decay 10^{-4}) with a one-cycle schedule, 40 epochs, batch 32 and horizontal flips; the epoch with the best validation Dice on real validation patients is kept and scored on the 74 real test patients. Dice is computed per patient over that patient’s slices and then averaged over patients; we report WT, TC, ET and their mean, as mean $\pm$ standard deviation over three seeds. Conditions: 10% (26 patients), 25% (64) and 100% (258) of the training patients, each with and without synthetic pairs, plus synthetic-only. The pool is the selected model’s 20,000-sample set conditioned on training masks, each sample paired with its mask. *Patient matching*: a real- $x\%$ segmenter only receives synthetic slices whose mask belongs to one of its own $x\%$ patients, so a low-data condition never sees tumour geometry of patients it does not have (about 1.25 synthetic slices per real slice). The real-only segmenters are also scored on all synthetic pairs (“mask consistency”: Dice between the prediction on the generated image and the conditioning mask).

ldm128_maskcond



ldm128_maskcond_reg

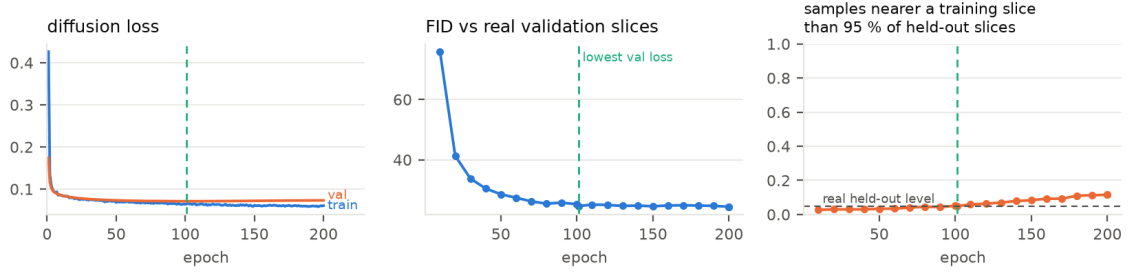


Figure 1: Validation diffusion loss, FID against real validation slices and memorised fraction per saved checkpoint for the unregularised 128px mask-conditioned model (top) and the regularised recipe (bottom). FID keeps improving after the validation-loss minimum in both, but the unregularised model does so by copying: 97% of samples are closer to a training slice than real held-out slices are by epoch 200.

Secondary analyses (declared after the first seed-0 results). (i) a 1:1 synthetic ratio; (ii) synthetic pre-training: 20 epochs on the synthetic pairs, then the standard 40 epochs on real data; (iii) VAE-reconstructed real: the real slices are replaced by their own reconstruction through the frozen autoencoder, with no generator involved, which measures the autoencoder’s share of any loss. A compute-matched control for (ii), declared after its first result, pre-trains for the same 20 epochs on the real slices themselves.

4.5 Boundary analysis: decoder fine-tuning

To test whether the autoencoder’s share is the decoder’s blur, we fine-tune only the decoder and its post-quantisation convolution (49.5M parameters) on the 15,895 training slices with $L_1 + 0.5 \cdot \text{LPIPS-VGG}$ per channel (AdamW 2×10^{-5} , batch 16, flips, 8 epochs, 57 min); the epoch with the lowest validation loss is kept. The encoder is untouched, so every cached latent and every trained diffusion model stay valid: the *same* latent samples are decoded again with the new decoder and re-scored, and the primary and secondary segmentation protocols are repeated with that pool, the VAE-reconstructed-real control now using the fine-tuned decoder. Two readings were fixed in advance: (a) the study stops if the reconstruction ceiling does not improve; (b) the loss is attributed to the decoder only if the VAE-reconstructed-real control moves towards real-only *and* real + synthetic moves the same way.

5 Results

The results follow the evaluation dimensions above: training behaviour and memorisation first, then held-out generation evidence, followed by the secondary segmentation assessment and analyses of its boundary conditions.

5.1 Training behaviour and memorisation risk

Figure 1 and Table 1 characterise how long training remains credible under the memorisation check. The baseline reaches its lowest validation loss at epoch 34; by epoch 200 its FID against real validation slices has fallen from 29.9 to 21.8, an improvement an FID-only selection rule would take, while 97% of its samples then lie closer to a training slice than 95% of real held-out slices do. Dropout alone changes little (0.891 memorised at epoch 200). Cached affine augmentation postpones the turn to epoch 101, lowers the memorised

Table 1: Checkpoint curves (2,000 samples per checkpoint, FID vs real validation slices). “Best” is the epoch with the lowest validation diffusion loss, the checkpoint used everywhere else. The selection rule (validation data only) picked the reg recipe.

model	best epoch	FID _{val}	KID $\times 10^3$	memorised	FID _{val} @200	memorised @200
LDM-128-mask (flips only)	34	29.89	18.18	0.086	21.81	0.966
+ dropout 0.1	41	28.82	17.24	0.093	21.99	0.891
+ dropout 0.1 + affine augmentation (reg, chosen)	101	24.65	12.86	0.049	24.43	0.115
LDM-128-reg (unconditional)	116	24.81	15.87 [†]	0.040	23.20	0.051
LDM-256-mask-reg	71	15.10	–	0.033	12.94	0.291

[†]KID of the unconditional model is against the test slices (Table 2); the curve records FID only.

Table 2: Generation quality against all 4,623 real test slices (5,000 samples; best-validation-loss EMA weights; DDIM 50). The FID between real validation and real test slices is 10.49 (128 px) and 9.55 (256 px); real pair-SSIM is 0.652 / 0.690. Per-modality FIDs replicate one grey channel to RGB and are far above the composite.

model	setting	FID	KID $\times 10^3$	FID FLAIR / T1ce / T2	pair-SSIM	memorised
LDM-128-mask (baseline)	$g=2$	26.22	20.89	45.9 / 45.4 / 73.0	0.576	0.069
LDM-128-mask + dropout	$g=2$	24.34	19.13	47.7 / 46.2 / 75.7	0.585	0.070
LDM-128-mask-reg	$g=2$	20.85	15.19	44.3 / 44.0 / 69.6	0.584	0.043
LDM-128-mask-reg	$g=1$	20.37	14.74	47.0 / 44.2 / 69.4	0.586	0.036
LDM-128-mask-reg	$g=2$, unseen masks	22.85	16.52	45.2 / 44.1 / 71.1	0.593	0.040
LDM-128-reg (unconditional)	$g=1$	21.31	15.87	47.5 / 45.0 / 68.7	0.579	0.027
LDM-256-mask-reg	$g=2$	10.45	6.40	40.3 / 39.6 / 59.1	0.628	0.030
LDM-256-mask-reg	$g=1$	12.83	8.77	43.1 / 41.7 / 58.2	0.637	0.021
LDM-256-mask-reg	$g=2$, unseen masks	10.83	6.17	40.4 / 39.3 / 60.7	0.633	0.024

fraction at the last epoch to 0.115 and gives the best early-stopped model (FID 24.7 vs 28.8 and 29.9). Even so, the regularised 256 px model starts copying after epoch ≈ 90 (0.074 at epoch 100, 0.291 at epoch 200): augmentation postpones memorisation, it does not remove it, and early stopping on validation loss remains necessary. All results below use the best-validation-loss checkpoint.

5.2 Distributional fidelity, diversity, and unseen-mask generation

Table 2 scores the early-stopped models on the held-out patients. The regularised recipe improves test FID by 5.4 points over the baseline at 128 px. Masks of validation patients, tumour shapes never seen in training, cost 2.0 FID points at 128 px and 0.4 at 256 px; together with the nearest-neighbour results, this is consistent with conditional generation beyond the training masks, although it is not an anatomical validation. Conditioning costs no fidelity (the mask-conditioned and unconditional 128 px models score alike), guidance 2 helps at 256 px and is marginally worse than guidance 1 at 128 px, and every selected checkpoint has a memorised fraction at or below the 0.05 expected of a model that generalises under this statistic. At 256 px the model reaches FID 10.45 against a floor of 9.55 between two sets of *real* slices (Fig. 2), encouraging distributional evidence within this dataset rather than proof of clinical realism. T2 is the hardest modality at both resolutions. The autoencoder ceiling is PSNR 26.4 dB / SSIM 0.833 / LPIPS 0.041 at 128 px and 29.0 dB / 0.877 / 0.036 at 256 px; every latent model remains bounded by this reconstruction stage.

5.3 Secondary assessment: segmentation utility at 128 px

Mixing the synthetic pairs into training did not improve the segmenter (Table 3, Fig. 3). Whole-tumour Dice rises slightly with 26 real patients (0.801 $\rightarrow$ 0.809), but TC and ET fall at every fraction (ET at 10 %: 0.509 $\rightarrow$ 0.430; mean Dice at 100 %: 0.778 $\rightarrow$ 0.757), and the loss is present in 9 of 9 seed-matched pairs. Synthetic data alone reaches 0.660 mean Dice, about the level of 26 real patients. Mask consistency helps localise where the loss sits: a real-trained segmenter recovers the whole-tumour outline of the synthetic images well (WT consistency 0.789 for the 100 % segmenter, vs 0.886 on real test slices) but agrees with the conditioning mask on ET only about half the time (0.494). The pairs retain the tumour outline but appear to weaken the TC/ET signal that the segmenter uses.

5.4 Secondary analyses of segmentation utility

Table 4 and Fig. 4 explore the three questions declared after the first seed-0 results. *Changing the ratio is insufficient*: capping synthetic slices at 1:1 recovers only about a third of the loss and ET stays 0.03–0.07 below

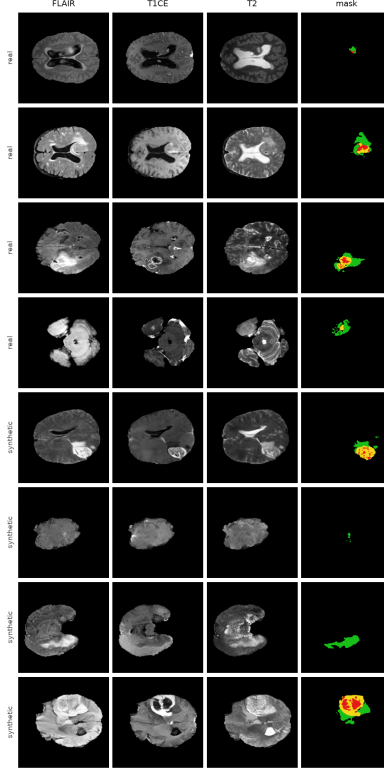


Figure 2: Real test slices (top four rows) and LDM-256-mask-reg samples (bottom four rows) with their masks. Colours: necrotic core, oedema, enhancing tumour.

Table 3: Per-patient test Dice at 128 px (74 test patients; mean $\pm$ std over 3 seeds). Synthetic pairs come from LDM-128-mask-reg and are patient-matched.

training data	patients	real slices	synthetic	WT	TC	ET	mean
10 % real	26	1,483–1,705	0	0.801 ± 0.011	0.580 ± 0.023	0.509 ± 0.022	0.630 ± 0.016
10 % real + synthetic	26	1,483–1,705	1,845–2,208	0.809 ± 0.006	0.559 ± 0.010	0.430 ± 0.033	0.600 ± 0.014
25 % real	64	3,937–4,031	0	0.847 ± 0.003	0.676 ± 0.011	0.611 ± 0.010	0.711 ± 0.004
25 % real + synthetic	64	3,937–4,031	4,913–5,158	0.844 ± 0.004	0.660 ± 0.021	0.521 ± 0.014	0.675 ± 0.012
100 % real	258	15,895	0	0.886 ± 0.001	0.775 ± 0.005	0.673 ± 0.004	0.778 ± 0.003
100 % real + synthetic	258	15,895	20,000	0.879 ± 0.001	0.742 ± 0.009	0.650 ± 0.010	0.757 ± 0.006
synthetic only	0	0	20,000	0.829 ± 0.002	0.669 ± 0.007	0.481 ± 0.012	0.660 ± 0.005

real-only. *The frozen autoencoder reproduces the loss without a generator:* replacing the real slices by their own reconstruction through `sd-vae-ft-mse` costs -0.057 / -0.072 / -0.072 mean Dice, again concentrated on ET (-0.11 to -0.14). Because every synthetic image passes through this autoencoder, the result is consistent with important enhancing-tumour information being lost before diffusion sampling. *Pre-training helps where mixing does not:* 20 epochs on synthetic pairs followed by the standard schedule on real data gains $+0.044$ mean Dice with 26 patients (all three regions, ET $0.509 \rightarrow 0.564$, 3/3 seeds) and $+0.015$ with 64, nothing with 258. Because the pre-trained segmenter sees 20 extra epochs, a control was declared that spends those epochs on the real slices: it recovers about half of the 10 % gain and none of the 25 % gain, leaving a synthetic-specific effect of $+0.021$ and $+0.014$ (3/3 seeds each). This pattern suggests that real-data fine-tuning can override synthetic-image limitations while retaining some transferable signal.

5.5 Resolution as a boundary condition: segmentation at 256 px

The primary protocol was repeated unchanged at 256 px, declared before any 256 px segmenter was trained (Table 5). With 26 real patients the sign flips: the patient-matched synthetic pairs improve every region (TC $+0.049$, ET $+0.022$, WT $+0.011$; 3/3 seeds, spread ≤ 0.004), where the same protocol at 128 px lost 0.030. With 64 patients the effect is neutral and with 258 it is neutral on the mean and still slightly negative on ET (-0.015). The synthetic-only segmenter reaches 0.760, the level of 25 % real data (128 px: 0.660, the level of 10 %). The 256 px pool comes from a generator at the FID floor decoded through an autoencoder whose

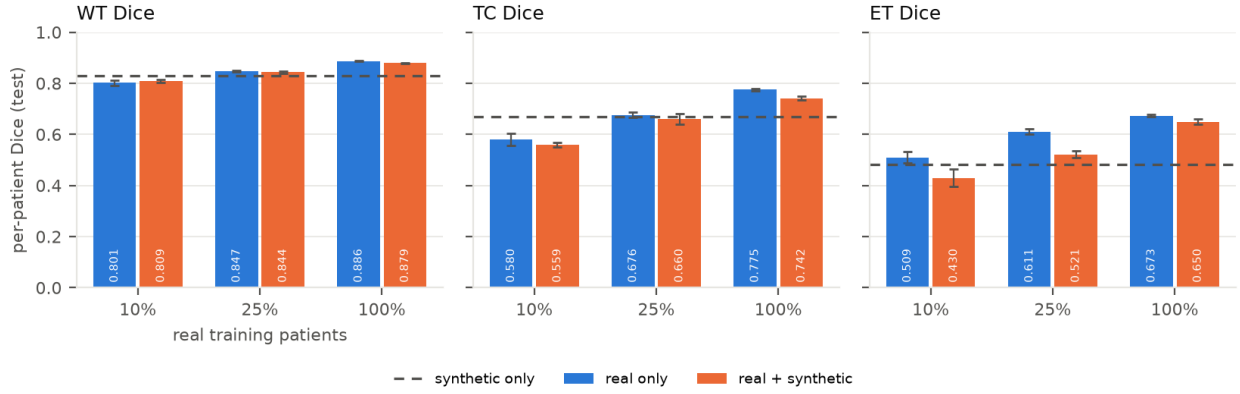


Figure 3: 128 px primary protocol: per-patient test Dice with and without patient-matched synthetic pairs, three seeds. The dashed line is the synthetic-only segmenter.

Table 4: Secondary analyses at 128 px: change in mean test Dice relative to real only, seed-matched, three seeds (per-region changes for the two decisive rows). The control row pre-trains for the same 20 extra epochs on the real slices.

condition	10 % real	25 % real	100 % real
real + synthetic, 1.25:1 (primary)	−0.030	−0.036	−0.021
real + synthetic, 1:1	−0.020	−0.024	−0.018
real replaced by its frozen-VAE reconstruction (no generator)	−0.057 (ET −0.112)	−0.072 (ET −0.135)	−0.072 (ET −0.131)
synthetic pre-training, then real	+0.044	+0.015	+0.001
real pre-training, then real (compute-matched control)	+0.023	+0.001	+0.004
synthetic pre-training – control	+0.021 (3/3 seeds)	+0.014 (3/3 seeds)	−0.003

ceiling is 2.6 dB higher; the real-only segmenters are stronger at 256 px as well, so the two resolutions are compared only within themselves.

5.6 Autoencoder as a boundary condition: decoder fine-tuning

Fine-tuning the decoder on the training slices improves every image metric (Table 6, Fig. 5): the ceiling rises from 26.37 to 27.79 dB (SSIM 0.833 → 0.883, LPIPS 0.041 → 0.032; T2, the worst channel, gains 1.6 dB), so reading (a) passed and the study continued. Decoding the same 20,000 latents again leaves the three-channel FID unchanged (20.85 → 20.89) but cuts the per-modality FIDs by 35–75 % (T2 69.6 → 17.6); the per-channel scores of the frozen decoder were dominated by the texture it invents, which the composite metric, seeing the three channels as one colour image, largely ignores. Sample diversity moves towards that of real slices (pair-SSIM 0.584 → 0.636, real 0.652), and the memorised fraction rises from 0.043 to 0.051 because every sample moves closer to every real slice under a sharper decoding; the latents are identical, so this is not new copying.

The segmenter, however, barely notices. With the re-decoded pool, mixing still costs Dice at every fraction and in 9 of 9 seed-matched pairs (−0.020 / −0.033 / −0.012; ET still −0.06 to −0.08), the re-trained real-only baselines reproduce the earlier ones within the seed spread, and the synthetic-only segmenter improves from 0.660 to 0.672. The VAE-reconstructed-real control moves towards real-only at every fraction (−0.057 / −0.072 / −0.072 → −0.044 / −0.055 / −0.045) and real + synthetic moves the same way, so the pre-declared condition (b) is met, but only in part: a decoder trained on these very slices still loses 0.044–0.055 mean Dice (ET 0.08–0.11) when real slices are merely passed through the autoencoder, which remains more than mixing costs. Within this setup, the controls are consistent with the autoencoder being the principal limitation and with most of that limitation arising in the encoder’s $16 \times 16 \times 4$ latent rather than the decoder; the decoder accounts for about a third of the observed loss. The pre-training gain is unchanged by the decoder (+0.018 / +0.015 against the control).

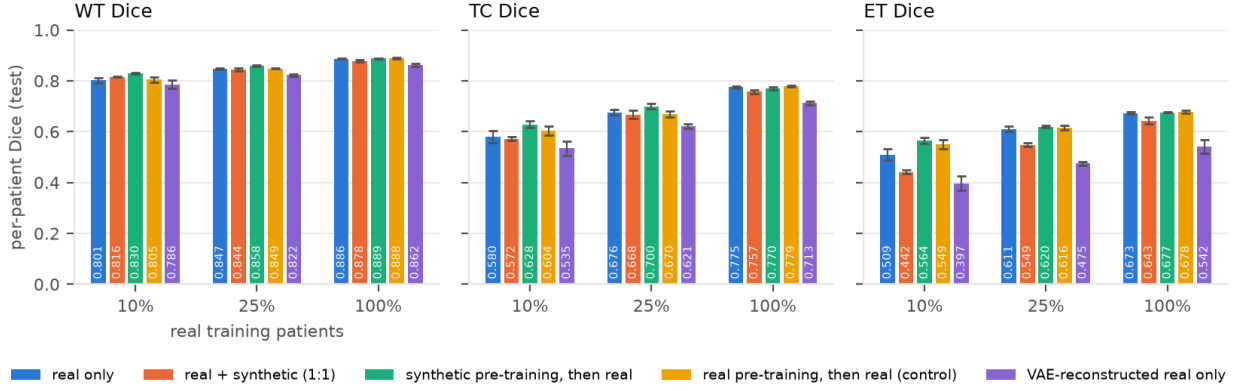


Figure 4: 128 px secondary analyses (frozen decoder): real only, real + synthetic 1:1, synthetic pre-training, real pre-training (control) and real slices reconstructed through the frozen autoencoder.

Table 5: Per-patient test Dice at 256 px with the LDM-256-mask-reg pool (same protocol, seeds and patient matching as Table 3).

training data	WT	TC	ET	mean	Δ mean
10 % real	0.849 ± 0.006	0.672 ± 0.019	0.604 ± 0.009	0.709 ± 0.003	
10 % real + synthetic	0.860 ± 0.006	0.721 ± 0.007	0.626 ± 0.006	0.736 ± 0.004	+0.027
25 % real	0.874 ± 0.003	0.751 ± 0.009	0.668 ± 0.010	0.764 ± 0.007	
25 % real + synthetic	0.878 ± 0.000	0.769 ± 0.004	0.654 ± 0.000	0.767 ± 0.001	+0.003
100 % real	0.895 ± 0.000	0.810 ± 0.002	0.707 ± 0.002	0.804 ± 0.000	
100 % real + synthetic	0.894 ± 0.004	0.818 ± 0.006	0.692 ± 0.008	0.801 ± 0.005	-0.003
synthetic only	0.871 ± 0.002	0.785 ± 0.008	0.623 ± 0.004	0.760 ± 0.003	

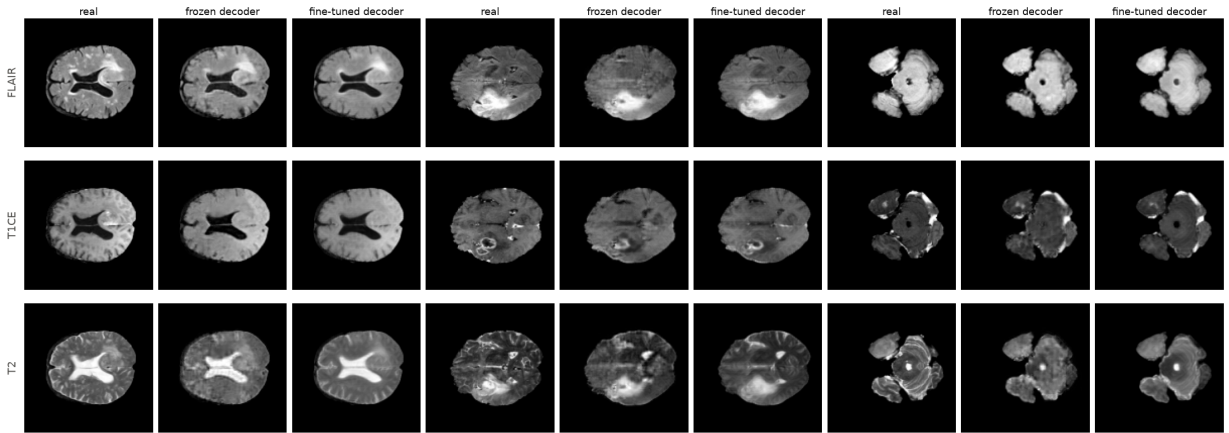


Figure 5: Three test slices with enhancing tumour reconstructed through the frozen and the fine-tuned decoder (same encoder, same latents). The blotchy texture the frozen decoder adds to T2 disappears and enhancing rims in T1ce come out sharper.

Table 6: Decoder fine-tuning at 128 px. Top: reconstruction ceiling on the 4,623 test slices and scores of the *same* 20,000 latent samples decoded with each decoder. Bottom: change in mean test Dice relative to real only (seed-matched, three seeds) when the segmentation protocols are repeated with the re-decoded pool.

	PSNR	SSIM	LPIPS	FID (3-ch)	FID FLAIR / T1ce / T2	memorised
frozen decoder	26.37 dB	0.833	0.041	20.85	44.3 / 44.0 / 69.6	0.043
fine-tuned decoder	27.79 dB	0.883	0.032	20.89	28.6 / 25.6 / 17.6	0.051
Δ mean Dice vs real only	10 % real		25 % real		100 % real	
	frozen	fine-tuned	frozen	fine-tuned	frozen	fine-tuned
real + synthetic (1.25:1)	−0.031	−0.020	−0.036	−0.033	−0.021	−0.012
ET only	−0.079	−0.064	−0.089	−0.081	−0.024	−0.022
real + synthetic (1:1)	−0.020	−0.007	−0.024	−0.025	−0.018	−0.012
real replaced by its VAE reconstruction	−0.057	−0.044	−0.072	−0.055	−0.072	−0.045
ET only	−0.112	−0.080	−0.135	−0.107	−0.131	−0.089
synthetic pre-training – control	+0.021	+0.018	+0.014	+0.015	−0.003	−0.001
synthetic only (mean Dice)	0.660 → 0.672					

6 Discussion

Viability assessment. The results support a limited and conditional account of viability rather than a general claim that latent diffusion solves synthetic brain MRI generation. Training and sampling are technically feasible in the tested 2-D setting, and the 256 px model provides encouraging distributional evidence relative to held-out data. That evidence depends on regularisation and validation-based early stopping: prolonged training improves FID while increasingly reproducing the training set. Generation from unseen-patient masks adds preliminary same-dataset evidence of conditional generalisation, while the segmentation experiments show that practical utility depends on resolution and on how the synthetic pairs are used.

Table 7: What the exploratory study does and does not establish. “Supported” is restricted to the reported 2-D BraTS 2020 setting.

dimension	evidence	finding	interpretation
Technical feasibility	Training and sampling at 128 and 256 px	Both resolutions produced reproducible model outputs	Supported for the tested pipeline
Distributional fidelity	FID/KID and a real-vs-real reference	256 px FID 10.45 vs a 9.55 reference	Encouraging metric-based evidence
Diversity and non-copying	Pair-SSIM and nearest-neighbour fraction	Selected checkpoints remain near the held-out reference; late checkpoints copy	Conditional on augmentation and early stopping
Unseen-mask generation	5,000 samples from validation-patient masks	Small FID change relative to training-patient masks	Preliminary support within the same dataset
Segmentation utility	Patient-matched, three-seed experiments	128 px mixing hurts; low-data pre-training and 256 px mixing can help	Conditional rather than general utility
Clinical validity	No radiologist study or external cohort	Not measured	Not established

What the results suggest. A latent diffusion model built on a natural-image autoencoder can produce BraTS slices with strong held-out fidelity scores while still developing a serious memorisation risk during prolonged training. At 128 px, the no-generator and decoder controls are consistent with the autoencoder bottleneck weakening the enhancing-tumour information needed for segmentation; improved image metrics after decoder fine-tuning do not remove the downstream loss. Two settings provide evidence of conditional utility: synthetic pre-training followed by real-data fine-tuning, and mixing at 256 px in the low-data regime. These are boundary observations, not evidence that synthetic augmentation will help across datasets, architectures or clinical tasks.

Implications for evaluation. Three practices were decisive and are cheap: early stopping on the validation diffusion loss with a memorisation check, since FID alone would have chosen the most memorised checkpoints; a no-generator control (real slices through the autoencoder), which localised the loss; and a compute-matched control for pre-training, which halved the apparent gain. Per-modality FIDs and the composite FID also disagree sharply once the decoder changes, so a single FID number is not a safe summary for multi-channel medical images.

Limitations and what remains untested. All models are 2-D and Dice is aggregated per patient over slices, not on volumes. FID/KID use Inception features; the real val-vs-test floor calibrates them, but a radiology-specific extractor [20] would be more sensitive to clinically relevant detail. No radiologist evaluated anatomical or diagnostic plausibility, and there is no privacy analysis or external test cohort. The encoder of the autoencoder is frozen; a medical-image autoencoder or an encoder fine-tune with the diffusion models retrained is the natural next experiment, and the present controls suggest it may matter more than decoder-only fine-tuning. The segmenter is a plain 2-D U-Net rather than a tuned nnU-Net [15], and the evidence comes from one dataset, one patient split and three segmentation seeds. The study therefore motivates further 3-D, external-dataset and expert-reader evaluation rather than clinical use.

7 Conclusion

Within the tested 2-D BraTS 2020 setting, latent diffusion is a plausible research pipeline for synthetic brain MRI generation, but its viability is conditional. Cached affine augmentation and validation-based early stopping are needed to control memorisation; higher resolution produces substantially stronger distributional evidence; and downstream value depends on whether the samples are mixed directly or used for pre-training before real-data fine-tuning. The controls suggest that the pretrained autoencoder, particularly its compressed latent representation at 128 px, is an important boundary on useful detail. These results justify further study of medical-image autoencoders, 3-D generation, external datasets and expert assessment, but they do not establish broad segmentation benefit or clinical validity. We release the code, patient split, pre-declared protocol and every run’s configuration and metrics so that this exploratory assessment can be reproduced and extended.

Reproducibility. Code, split file, protocol declarations with timestamps, result tables and figures: <https://github.com/wasiqnauman/SynthMRI>. One command reproduces the full experiment set on a single 16 GB GPU in about 40 hours; every reported number is traceable to a run directory that records the resolved configuration, git commit and metrics.

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